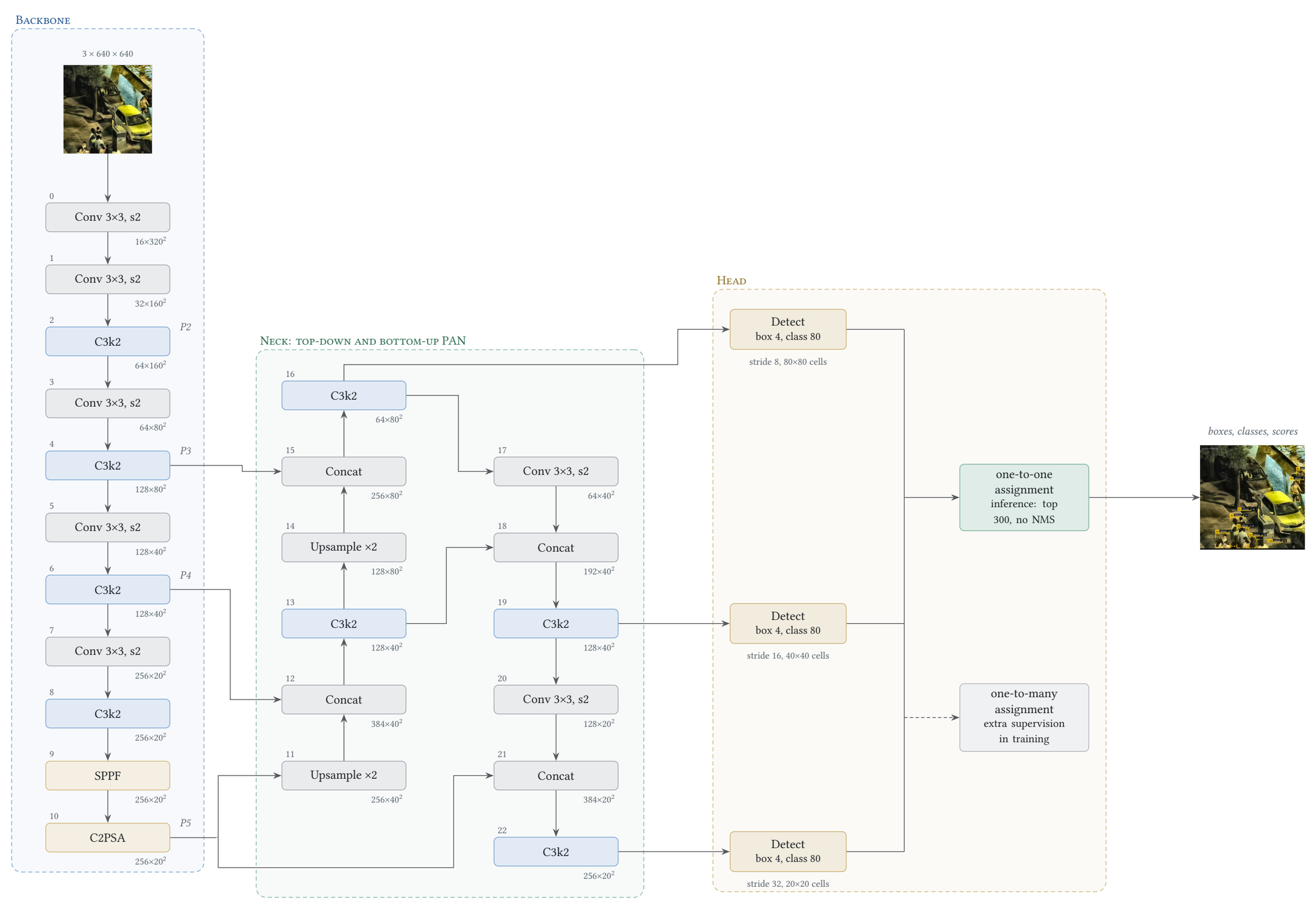

Backbone
3 × 640 × 640
0
Conv 3×3, s2
16×320²
1
Conv 3×3, s2
32×160²
2
C3k2
P2
64×160²
3
Conv 3×3, s2
64×80²
4
C3k2
P3
128×80²
5
Conv 3×3, s2
128×40²
6
C3k2
P4
128×40²
7
Conv 3×3, s2
256×20²
8
C3k2
256×20²
9
SPPF
256×20²
10
C2PSA
P5
256×20²
Neck: top-down and bottom-up PAN
16
C3k2
64×80²
15
Concat
256×80²
14
Upsample ×2
128×80²
13
C3k2
128×40²
12
Concat
384×40²
11
Upsample ×2
256×40²
17
Conv 3×3, s2
64×40²
18
Concat
192×40²
19
C3k2
128×40²
20
Conv 3×3, s2
128×20²
21
Concat
384×20²
22
C3k2
256×20²
Head
Detect
box 4, class 80
stride 8, 80×80 cells
Detect
box 4, class 80
stride 16, 40×40 cells
Detect
box 4, class 80
stride 32, 20×20 cells
one-to-one
assignment
inference: top
300, no NMS
one-to-many
assignment
extra supervision
in training
boxes, classes, scores